

SCHOOL OF COMPUTING SCIENCE & ENGINEERING

PROJECT APPROVAL FORM AND ABSTRACT

Odd 2025-2026

B. Tech Project Details

Project ID:

Title	SNN-Based Malware Classification Using Program Execution Trace.		
Project Type	☐ Community based design problem (Interdisciplinary) ☐ App Development / Utility ☐ IOT/Hardware based ☒ AI/ML/Data Science ☐ Healthcare Projects	Project Outcome	☒ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG Goal Number	
Publication Target	☐ SCOPUS Journal ☐ SCOPUS Conference ☐ SCOPUS Book Chapter ☒ SCI Journal	Guide Name: Dr. Santosh Kumar	

Student Details:

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Sustainable Development Goals (SDG Goals), Targets and Indicators

- Goal 1: End poverty in all its forms everywhere
- Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.
- Goal 3: Ensure healthy lives and promote well-being for all at all ages
- Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
- Goal 5: Achieve gender equality and empower all women and girls
- Goal 6: Ensure availability and sustainable management of water and sanitation for all
- Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all
- Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
- Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
- Goal 10: Reduce inequality within and among countries
- Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable
- Goal 12: Ensure sustainable consumption and production patterns
- Goal 13: Take urgent action to combat climate change and its impacts
- Goal 14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
- Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
- Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
- Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

SNN – Based Malware Classification Using Program Execution Trace

Area/Domain of Project:

Cyber – Security & AI/ML/Data Science

Abstract:

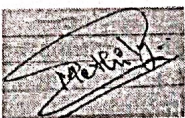
Modern malware is becoming smarter and harder to detect, creating a major problem for today's cybersecurity systems. Standard antivirus tools often fail to catch brand-new threats or malware that constantly changes its code to avoid being found. This research introduces a new way to classify malware using a brain-inspired technology called a Spiking Neural Network (SNN). Instead of just scanning files, our method analyzes a program's behavior by tracking its actions step-by-step as it runs. This creates a detailed "behavioral fingerprint" for any piece of software.

We convert this fingerprint into a pattern of timed signals that SNNs are uniquely designed to understand. The SNN then learns to recognize the subtle, time-based patterns that separate safe software from malicious activity. This approach uses the natural strengths of SNNs—which are fast, energy-efficient, and excellent at understanding sequences of events—to build a strong and reliable detection system. The result is a powerful new tool designed to accurately identify harmful software and provide a more advanced defense against complex cyber threats.

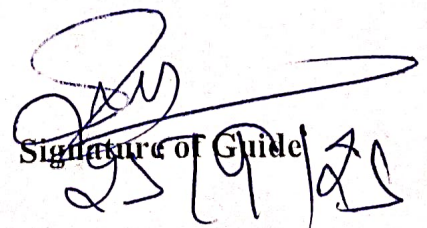
Problem Statement:

The increasing sophistication of malware presents a critical and ongoing threat to cybersecurity. Conventional malware detection systems, which primarily rely on signature-based scanning, are proving insufficient. These methods are fundamentally reactive, as they can only identify threats for which a signature already exists. Consequently, they are often ineffective against zero-day exploits and polymorphic or metamorphic malware, which dynamically alter their code to evade detection.

While traditional machine learning models have improved detection rates, they can still be bypassed by advanced evasion techniques and often struggle to interpret the complex, time-dependent behaviors that characterize modern threats. This creates a significant security gap where novel and adaptive malware can operate undetected, leading to data breaches, system compromises, and financial loss. Therefore, the central problem is the urgent need for a new detection paradigm that moves beyond static signatures to analyze the intrinsic runtime behavior of programs, enabling the identification of malicious software based on its actions rather than its appearance.



Signature of Students



Signature of Guide